

Doing things graphically

GUI tools are more convenient for certain use cases. Debian includes many ways to manage packages. We will look at each of them. Let's start with the famous *aptitude*.

aptitude is a semi-graphical package management software, which comes with the power of the command line as well. To start *aptitude*, issue the following command:

```
aptitude
```

You'll notice something similar to the image in Figure 1. Although, it doesn't look very user friendly, it has lots of power. For instance, it can effectively scan your system and give you suggestions to remove some useless packages from your system.

Synaptic is a powerful graphical way to manage your packages. To start Synaptic, go to **System** → **Administration** → **Synaptic Package Manager**. Synaptic, as we all know, is easy to use. Pressing the 'reload' button will implement an *apt-get update*, while 'mark all updates' is the same as an *apt-get upgrade*. You can also manage offline installation with CDs/DVDs, using Synaptic. Go to **Edit** → **Add CD-ROM**. You can also manage the online software repositories with it. As you can see in Figure 2, it will even display a screenshot of the application with details on what the selected package does.

Update-manager is yet another graphical utility to manage easy package updates. After you log in to your desktop, by default, it runs in the background to check if your system packages are up to date or not. If there are updates, you get a GUI notification near your task bar.

When things go wrong

We can't really say when things will go wrong, but they sometimes do, nonetheless. Apt has some options to correct corrupt installations. To get things in order, issue the following command:

```
apt-get -f install
```

Note that there is no argument after 'install'. It will automatically detect half-installed or broken packages and correct them.

You can find the status of all packages using Synaptic or by using the following command afterwards:

```
dpkg -l
```

The above command will give an output of the list of installed packages.

You can also view the list of files (including their locations) installed by a particular package, using the following command:

```
dpkg -L package_name
```

Just in case it's absolutely impossible to rectify a corrupt installed package, you can forcefully remove it from your system, using the code below:

```
dpkg --force-all package_name
```

"I want the latest software in Lenny!"

Sure. You can get it!

There are different ways to do so. The first option is to determine your requirements. Do you want only some software? Or do you want to use the latest software without going to 'unstable' or 'experimental' and take the risk of breaking your current stable working set-up?

Debian has a project called backports. This is recompiled software from 'testing' or 'unstable' branches for the 'stable' version. To get backports of software, simply add the following lines into your */etc/apt/sources.list* file:

```
deb http://www.backports.org/debian lenny-backports main contrib non-free
```

Run *apt-get update* and install a more recent version of your favourite application. For more information, visit backports.org.

Another way is to use 'pinning', which is a way to give higher priority to particular repositories for particular package(s). For example, if you have added the 'experimental' repository in your */etc/apt/sources.list* and want to install only 'mutt' from it, you can add the following lines into */etc/apt/preferences*; create this file, if it is not already there:

```
Package: mutt
```

```
Pin: release a=experimental
```

```
Pin-Priority: 999
```

Now, when you execute *apt-get install mutt* it will fetch 'mutt' from 'experimental'.

We have gone over some basics of Debian packages here. You can always visit www.debian.org for more details. There are some books available too, for example, *Debian 3.1 GNU/Linux Bible* by Jaldhar Vyas, or *The Debian System* by Martin Krafft. The best way to learn more about Debian is to install Debian and use it, practically :-)



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